

We modeling penalty corners in field hockey as a game between the offense and defense. At each corner, the offense chooses a setup and strategy with the objective of scoring, while the defense chooses a strategy to prevent a goal. Corners are generated by fouls committed inside the circle, so the number of corners in a match is not known in advance. In particular, any corner may be the final corner of the match, but teams do not know *ex ante* when the final corner will occur.

We therefore model corners as an indefinite horizon zero sum repeated game in which each corner constitutes a stage game. Each stage game has two parts: teams first observe the current state of the match, and then simultaneously choose offensive and defensive strategies.

$$\text{For each corner } t : \left\{ \begin{array}{l} S_t = (H_t, \text{score}_t, \text{time}_t, F_t), \\ \sigma^O(\cdot | S_t) \in \Delta(\mathcal{A}^O), \quad \sigma^D(\cdot | S_t) \in \Delta(\mathcal{A}^D), \\ A_t^O \sim \sigma^O(\cdot | S_t), \quad A_t^D \sim \sigma^D(\cdot | S_t), \\ Y_t \in \{\text{Goal}, \text{No Goal}\}, \\ H_{t+1} = (H_t, A_t^O, A_t^D, Y_t), \\ S_{t+1} = (H_{t+1}, \text{score}_{t+1}, \text{time}_{t+1}, F_{t+1}). \end{array} \right.$$

where

- S_t is the state of the game at corner t , composed of the relevant corner history, the game score, the game time, and the observed APC formation.
- $\mathcal{A}^O$ and $\mathcal{A}^D$ are the sets of possible offensive and defensive actions, respectively.
- $\sigma^O(\cdot | S_t)$ and $\sigma^D(\cdot | S_t)$ are the offensive and defensive strategies conditional on the state S_t . Each strategy specifies a probability distribution over the relevant action set.
- A_t^O and A_t^D are the realized offensive and defensive actions drawn from these strategies.
- Y_t is the realized outcome of the corner.
- S_{t+1} is the state at the next corner. Thus, after each corner, the state is updated to incorporate the new information generated by the current corner.

Stage Game

Part 1: Common information is revealed. Once teams know that a penalty corner will occur, they observe the relevant game state, including the time remaining, score, APC initial setup, and other publicly available information.

Part 2: Strategies are chosen simultaneously. Conditional on this information, the offense and defense simultaneously choose their strategies. The resulting outcome is then realized (e.g., goal or no goal).

We know that a mixed strategy Nash equilibrium exists in this game for each state S_t by Nash's existence theorem. Thus we can calculate best response correspondences for each team.

Proof of Concept/Simple Example:

Consider some state S_t .

The offense has strategies: $\sigma_O(\cdot \mid S_t)$ where their action A_t^O will be drawn from.

The defense has strategies: $\sigma_D(\cdot \mid S_t)$ where their action A_t^D is drawn from.

The offense maximizes:

$$u^O(A_t^O, A_t^D, S_t) = P(Y_t = \text{Goal} \mid A_t^O, A_t^D, S_t)$$

Since the game is zero sum, the defense gets:

$$u^D(A_t^O, A_t^D, S_t) = -u^O$$

Using existing data, we have some payoff matrix that we can populate for each S_t :

	A_1^O	A_2^O	A_3^O	$\dots$	A_n^O
A_1^D					
A_2^D					
A_3^D					
$\vdots$					
A_h^D					

This is what we need to calculate with data. Each entry is just

$$P(\text{Goal} \mid A_t^D, A_t^O, S_t).$$

For example, with two strategies, we might have the following payoff matrix:

	A_1^O	A_2^O
A_1^D	0.5	0.5
A_2^D	0.75	0.25

which would represent the following game, where offense maximizes the probability of

a goal and defense minimizes it. Equivalently, if defense utility is defined as the negative probability of a goal, both players maximize their own utility:

	A_1^O	A_2^O
A_1^D	(-.5, .5)	(-.5, .5)
A_2^D	(-.75, .75)	(-.25, .25)

The upper-left cell is a Nash equilibrium. Given that the offense plays A_1^O , the defense prefers A_1^D , since

$$-.5 > -.75.$$

Given that the defense plays A_1^D , the offense is indifferent between A_1^O and A_2^O , since both give payoff .5. Thus neither player has a profitable unilateral deviation from (A_1^D, A_1^O) .

We could also have a game like this:

	A_1^O	A_2^O
A_1^D	(-.25, .25)	(-.75, .75)
A_2^D	(-.75, .75)	(-.25, .25)

Here, there is no pure strategy Nash equilibrium.

Let $p = \Pr(A_1^O)$ and $q = \Pr(A_1^D)$. We look for indifference point:

$$0.75p + 0.25(1 - p) = .25p + 0.75(1 - p)$$

$$p = 1/2$$

and

$$0.25q + 0.75(1 - q) = 0.75q + 0.25(1 - q)$$

$$q = 1/2$$

Thus the Nash is $p = 1/2, q = 1/2$.

When we have a larger matrix, we can eliminate any dominated strategies and then solve for a Nash equilibrium. The resulting equilibrium will specify the probability with which each team should play each strategy in its equilibrium support.

Open Questions:

- How should history be populated?

- Could be related to just this game, to past games/scouting, to team practices with certain corners and so on.
- Are there other variables we can include to make S_t more realistic? Anything we can remove? How granular can we make it?